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RESEARCH

The Coming

AI & Humanoid Robot Economic Shock

Economic, Fiscal, and Asset-Pricing Implications, 2026-2036

Institutional Investor Edition

Coupled-feedback macroeconomic simulation covering 26 entities, 100% of the global labor force

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AI usage calibration: Anthropic Economic Index (2025)

Executive Summary

This report presents the findings of a coupled-feedback macroeconomic simulation projecting the impact of artificial intelligence (AI) and humanoid robots on 26 national economies (100% of global labor force) over a 10-year horizon. The simulation combines deterministic modelling with 1,000-run Monte Carlo perturbation across 11 policy scenarios (11,000 total trajectories), cross-audited by three independent large language models (GPT-4o, Grok-3, Gemini 2.5).

The core finding: even under central-case assumptions, asset markets currently appear to be pricing an economic path that only has approximately 15-20% probability. The tail risks — mass unemployment, sovereign default cascade, developing-world migration crisis — are substantial and not reflected in current asset prices.

The November 3, 2026 US midterm election is the single most important near-term event for this probability distribution. A decisive opposition mandate would shift the probability of a managed transition or smooth landing to approximately 20-30%; a status-quo outcome would hold it at 8-12%. The current administration's January 2025 revocation of Executive Order 14110 on Safe, Secure, and Trustworthy Development of Artificial Intelligence, its stated opposition to Universal Basic Income, its deficit-busting policies of tax cuts coupled with increased spending, and its deregulation-first framework are incompatible with the proactive policy package required for any managed transition. If a Congressional mandate does not successfully materialize in 2026, action slips to the 2028 presidential election — by which point reactive policy arrives too late to prevent the very negative scenarios.

Headline Findings

- **Economic displacement scale.** Job displacement is projected to range from 250 million (Middle Path) to 1.69 billion jobs globally over 10 years. Critically, new job creation is unlikely to be significant or rapid enough to buffer these impacts. The Middle Path broadly aligns with mainstream forecasts; the upper range reflects cheap humanoid robots, widespread self-driving vehicle transformation, and white-collar AI job cuts hitting the labor market simultaneously.
- **Augmentation precedes automation.** The Anthropic Economic Index currently tracks Claude usage at 57% augmentation versus 43% automation. Augmentation reduces hiring before eliminating existing workers, meaning recent-graduate unemployment rises first as entry-level positions disappear. This serves as a leading indicator, appearing 18-24 months ahead of any significant aggregate unemployment rate (UR) rise.

- **Developing-world wage-floor trap.** Humanoid robot effective costs are projected to decline from approximately $18 per hour in 2026 to $0.30-$1.00 per hour by 2035, driven by Wright's Law with learning rates of 20-40% per doubling of cumulative production. Consequently, every existing human wage tranche — from Nigerian informal labor at approximately $0.45 per hour to Indian formal work at approximately $1.60 per hour — becomes uncompetitive by 2035-2040. This effectively closes the historical development ladder (Japan to Korea to China to Vietnam to Bangladesh).
- **Remittance reversal is significantly underpriced.** Current global remittance flows, approximately $700 billion per year (World Bank 2024), originate primarily from service-sector and gig jobs held by migrant workers in developed economies — precisely the jobs most exposed to AI and humanoid robot automation. In the Full Collapse scenario, these flows are projected to compress by 60-85% over 10 years. Countries where remittances exceed 10% of GDP — such as Honduras (22%), Nepal (22%), El Salvador (22%), Tajikistan (27%), and Tonga (37%) — face simultaneous fiscal and household income collapses. No emerging-market sovereign spread currently prices this scenario correctly.
- **Sovereign default cascade likely.** Base-case probability of any major sovereign default over 10 years: 40-80%, concentrated in developing economies with weak fiscal capacity (sub-7% tax-to-GDP) — Pakistan, Nigeria, Egypt, Bangladesh, Sri Lanka, Ethiopia. Default synchronicity risk is high because displacement affects them simultaneously, unlike serial defaults of past cycles.

Asset Pricing Implications — Central Thesis

Current asset prices reflect the mainstream 'consensus' path as if it were certain. However, our simulation distribution suggests this path has only a 15-25% probability under realistic policy-response assumptions.

Asset	Current (approx.)	Model-consistent (10y)	Mispricing direction
US 10-year Treasury	~4.5% yield	6.5-9% given probability-weighted tail distribution	TOO LOW by ~200-450 bps
Investment-grade credit spreads	~100 bps	300-500 bps (late-2008 analog)	TOO TIGHT by ~200-400 bps
S&P 500 (cap-weighted) forward P/E	~21	Sustainable 20-25 on AI-winner concentration	Approximately fair
S&P 500 equal-weighted forward P/E	~18	Compresses to 12-14 under displacement	OVERVALUED by ~30%
Russell 2000 forward P/E	~25	Compresses to 15-18 (consumer-discretionary exposure)	OVERVALUED by ~35%

Asset	Current (approx.)	Model-consistent (10y)	Mispricing direction
VIX (equity vol)	~15-18	25-35 (sustained tail regime)	TOO LOW by ~50-100%
Gold (crisis-insurance premium, not intrinsic value)	~$3,200 per oz	$4,500-6,000 per oz (probability-weighted; see scenario table)	UNDERPRICED by 30-80%
EM sovereign debt	10-15% yields on stressed names	20-30%+ given 80%+ 10y default probability	CATASTROPHICALLY MISPRICED
Long-duration Treasuries (30y)	~4.8%	6-12% given inflation/default range	VERY MISPRICED
Remittance-exposed EM currencies	Current yields moderate	Higher default AND depreciation risk	MATERIALLY UNDERPRICED

The single most-mispriced asset class, in our view, is emerging-market (EM) sovereign debt. The simulation gives approximately 80% probability of at least one major sovereign default over 10 years, concentrated in remittance-dependent economies. Current EM spreads imply less than 5% ten-year default probability — an order of magnitude understatement.

Why Markets Aren't Pricing This

- **Narrative dominance:** AI is priced as mostly upside (productivity miracle, Nvidia-type equity logic), with essentially zero weight on labor-demand destruction that produces deflation and sovereign stress.
- **Career risk asymmetry:** Portfolio managers face immediate risk for missing the AI rally, minimal risk for mispricing a 5-10-year tail event.
- **Institutional mandate constraints:** Pension funds, insurers, endowments have duration and credit allocations locked by mandates; cannot reposition at scale.
- **Central bank put assumption:** Markets still assume Federal Reserve / European Central Bank / Bank of Japan backstop. At debt-to-GDP above 250% with deflation, conventional monetary tools lose effectiveness.
- **Horizon mismatch:** Markets price 1-5 years ahead. The scenarios play out over 5-10 years, outside most fundamental pricing windows.

Key Sensitivities

- If humanoid robot unit cost follows Wright's Law at 40%/year decline (robots-build-robots), displacement arrives 3-5 years earlier than our central case. This is the robots_make_robots scenario — treated in detail in the Counter-Argument section ('Capital-building-capital dynamics') and reflected in the probability-adjusted outcome table (10% prior weight).

- If proactive policy (Universal Basic Income or UBI; AI-profit tax; grid mobilization) materializes in the US in 2027, outcomes improve materially but probability of this happening is constrained by historical base rates of coordinated policy (climate change analog).
- If developing-world sovereign defaults cluster synchronously, IMF-type multilateral capacity (approximately USD 1 trillion) is insufficient for the aggregate problem (estimated USD 5-10 trillion).
- If AI adoption rates match Anthropic Economic Index trajectory, white-collar displacement starts earlier than blue-collar — office/admin, computer/math, business/finance, legal are at 74-82% AI exposure today.

Acronyms and Terms

Term	Definition
AI	Artificial Intelligence — software that performs tasks typically requiring human cognition (writing, analysis, coding, decision support).
AGI	Artificial General Intelligence — AI that matches or exceeds human performance across most cognitive tasks.
BLS	Bureau of Labor Statistics — US Department of Labor data source.
BPS	Basis points — 1 bps = 0.01%. Used for interest rate spreads.
CB	Central Bank — e.g., US Federal Reserve, European Central Bank (ECB), Bank of Japan (BoJ).
CBO	Congressional Budget Office — US federal agency providing budget analysis.
CPI	Consumer Price Index — standard inflation measure.
DXA	Document unit in Microsoft Word (1440 DXA = 1 inch). Used internally.
EM	Emerging Markets — developing economies' financial assets.
GDP	Gross Domestic Product — total value of goods and services produced in an economy.
GFC	Global Financial Crisis — 2007-2009 event.
IG	Investment-Grade — credit-rated BBB/Baa or higher; opposite of high-yield (HY).
IMF	International Monetary Fund — multilateral lender of last resort.
IFR	International Federation of Robotics — industry data source.
LFPR	Labor Force Participation Rate — percentage of working-age adults in the workforce.
LR	Learning Rate (Wright's Law) — fractional cost decline per doubling of cumulative production.
MPC	Marginal Propensity to Consume — fraction of a dollar of income that a household spends.
Monte Carlo	Probabilistic simulation technique — running many perturbed scenarios to build distributions.
P/E	Price-to-Earnings ratio — standard equity valuation metric.
SOC	Standard Occupational Classification — BLS occupation coding (2-digit used here).
TAM	Total Addressable Market — the full potential demand for a product or service.
TFP	Total Factor Productivity — growth in output not explained by labor and capital increases.
TWh	Terawatt-hours — 10^12 watt-hours; unit of energy used for annual electricity consumption.
UBI	Universal Basic Income — unconditional cash payment to every adult citizen.

Term	Definition
UR	Unemployment Rate — percentage of labor force actively seeking work but unemployed.
Wright's Law	Empirical observation that unit cost falls by a fixed percentage for each doubling of cumulative production.

Model Framework

The simulation is a deterministic, coupled state machine advancing in 6-month steps over a 10-year horizon. Each of 26 entities (25 countries plus a rest-of-world aggregate) maintains approximately 40 state variables covering labor markets, output, distribution, fiscal accounts, household debt, prices, grid capacity, and social indicators. At each step, every state variable is updated using the current values of every other state variable — no variable evolves in isolation. The Monte Carlo layer runs 1,000 perturbed trajectories per scenario to produce full probability distributions.

Core Feedback Loops

- Employment → tax receipts → primary deficit → government debt-to-GDP → interest rate → fiscal space (crowding out)
- Tax rate increases → labor supply → employment → tax base (Laffer-like response, Chetty-calibrated)
- Spending cuts → aggregate demand → GDP → tax base → sovereign debt dynamics (Keynesian multiplier ~0.8)
- UBI transfers → consumer spending (high marginal propensity to consume) → service-sector employment
- AI capacity expansion → supply shock → deflationary pressure → Fisher debt-deflation spiral → real interest rate
- Household displacement → default rates → bank credit losses → credit crunch → further employment decline
- Unrest → regulation intensity → deployment speed → displacement rate → unrest (stabilizing feedback)
- AI/robot tax rate → deployment speed → productivity contribution → AI value-added (tax-base self-limitation)
- Electricity demand vs grid supply → deployment speed cap
- Sovereign default trigger → nominal debt haircut + currency devaluation + austerity → reset and partial recovery
- Developing-world displacement + unrest → migration outflow → developed-world inflow → border-defense cost + mass-expulsion risk
- Remittance flow from developed-world service jobs → developing-world household income → developing-world fiscal revenue

Country Coverage

The 26 modeled entities cover approximately 3.63 billion workers — 100% of the global labor force:

- Developed: United States, United Kingdom, Germany, France, Japan, Italy, Spain, Canada, Australia, South Korea
- China (separate bucket due to state-directed regime)
- Developing: India, Brazil, Nigeria, Mexico, Indonesia, Vietnam, Philippines, Thailand, Pakistan, Bangladesh, Ethiopia, Egypt, Turkey, Russia
- Rest-of-World aggregate (approx. 900 million labor force)

AI Usage Calibration

Occupational AI-exposure coefficients are calibrated against the Anthropic Economic Index (2025), which tracks actual Claude usage intensity across US Standard Occupational Classification (SOC) 2-digit categories. Top exposure: Office and Administrative Support (0.82), Computer and Mathematical (0.82), Business and Financial Operations (0.76), Legal (0.74), Arts/Design/Media (0.70). Lowest: Building and Grounds (0.18), Protective Services (0.20), Personal Care (0.20), Construction (0.22).

Derivation of the $18/hour 2026 Humanoid Robot Cost Anchor

The starting effective cost anchor of USD 18/hour in 2026 is built bottom-up from the cost-of-ownership components disclosed by humanoid-manufacturer investor decks and independent sell-side research. The components:

Component	USD per productive hour (2026)
Hardware amortization — unit cost USD 30,000-50,000 at 2026 scale (Tesla Optimus target, Figure 02 contract pricing, Unitree commercial). 5-year straight-line at 22 h/day × 365 d duty cycle	USD 0.75 - 1.25 / hr
Electricity — 1 kW active draw × USD 0.08-0.15/kWh industrial-rate average	USD 0.08 - 0.15 / hr
Maintenance, parts, actuator replacement (industrial-robot benchmark 8-12% of hardware cost annually)	USD 0.40 - 0.65 / hr
Software/subscription/cloud inference for perception + planning	USD 1.00 - 2.00 / hr
Insurance, safety compliance, liability reserve	USD 0.50 - 1.00 / hr
Remote human supervisor — one supervisor per 20-50 robots at developed-world wages (still required in 2026)	USD 1.00 - 3.00 / hr
Integration, site preparation, workflow customization amortized	USD 1.00 - 2.00 / hr

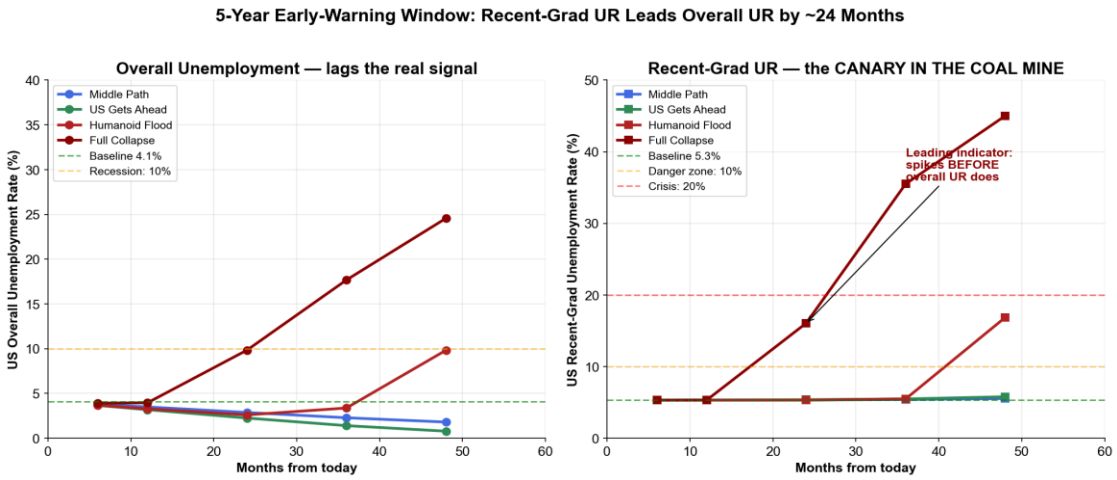
Component	USD per productive hour (2026)
Financing cost of capital @ 8-10% WACC on hardware + integration	USD 0.30 - 0.60 / hr
Risk / downtime / idle-capacity loading factor (typical industrial ~25-35%)	USD 1.50 - 3.00 / hr
Reliability and warranty reserve (first-generation premium)	USD 0.50 - 1.00 / hr
Sum (point estimate, middle of each band)	approximately USD 12-17 / hr
Sum (conservative, top of each band)	approximately USD 18-25 / hr

We anchor to USD 18/hour as a deliberately conservative upper-mid band for 2026 "effective" cost — meaning all-in cost to replace a human worker on the job, not just hardware acquisition amortized. This is consistent with the public ranges disclosed by Morgan Stanley humanoid research (2024-2025: USD 10-20/hr 2026 total cost of ownership), Goldman Sachs humanoid robotics note (January 2024: USD 15-20/hr mid-case), and ARK Invest Big Ideas (2024: USD 10-18/hr). Tesla's publicly stated target is approximately USD 20-30k unit cost at scale, which on hardware alone is USD 0.50-0.80/hr but does not include the supporting labor, integration, and software costs above.

The decline trajectory from USD 18/hr in 2026 follows scenario-specific learning rates — 20-40% per doubling of cumulative production (Wright's Law) — with the upper bound governed by the capital-building-capital dynamic (robots producing robots, AI generating AI agents) that pushes effective learning rates toward semiconductor-manufacturing territory (30-50% per doubling, compared with industrial-robot historical rates of ~10% per doubling).

Leading Indicators — Testable Predictions (Next 5 Years)

Aggregate unemployment rate (UR) lags the transition by 18-24 months. Earlier signals appear in recent-graduate labor markets, entry-level job openings, and cross-sector hiring patterns. We publish testable short-horizon predictions (3, 6, 12, and 18 months) and a 5-year watch-list to enable external validation. If our short-horizon predictions verify, the longer-horizon scenario probabilities tighten toward the severe branches; if they do not, the consensus path strengthens.



US recent-graduate UR (right) leads overall UR (left) by 18-24 months across scenarios. The gap is the leading-indicator signal.

Testable Predictions — Check Us at 3, 6, 12, 18 Months

Checkpoint	US recent-grad UR	Humanoid robot deployments global	Software new-grad postings	AI-cited layoffs (annual run-rate)
T+3 mo (Jul 2026)	5.5-5.8%	75-120k cumulative	Down 45-55% from 2022 peak	100-150k new-grad
T+6 mo (Oct 2026)	5.8-6.5%	120-220k cumulative	Down 55-65% from 2022 peak	150-220k new-grad
T+12 mo (Apr 2027)	6.5-8.0%	300-600k cumulative	Down 65-75% from 2022 peak	250-400k new-grad
T+18 mo (Oct 2027)	8.0-11.0%	600k-1.2M cumulative	Down 75-85% from 2022 peak	500k-1M new-grad

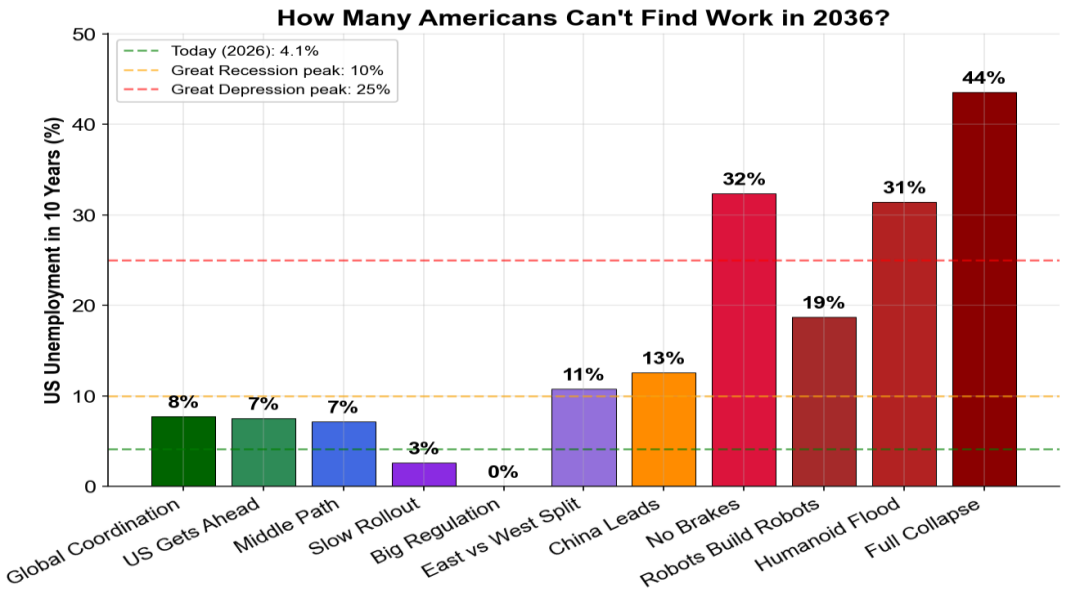
Primary Data Sources

Indicator	Source (authoritative, updated regularly)	URL
Recent-graduate unemployment	Federal Reserve Bank of New York — Labor Market for Recent College Graduates	newyorkfed.org/research/college-labor-market
Humanoid robot unit shipments	International Federation of Robotics (IFR) — World Robotics Report	ifr.org/world-robotics-reports
Company production guidance	Tesla, Figure AI, Agility Robotics, 1X, Unitree — quarterly earnings calls	investor relations sites (e.g., ir.tesla.com)
Software engineering job postings	LinkedIn Economic Graph; Indeed Hiring Lab; ADP Employment Report	economicgraph.linkedin.com; hiringlab.org; adpemploymentreport.com
AI-attributed layoffs	Challenger, Gray and Christmas — monthly layoff report	challengergray.com
AI usage intensity by occupation	Anthropic Economic Index	anthropic.com/news/economic-index
Corporate hiring disclosures	Fortune 500 earnings calls; SEC 8-K filings	sec.gov/edgar; individual IR pages

Scenarios & Probability Distributions

Eleven named scenarios span the plausible range of AI-labor impact futures. Each has been run 1,000 times via Monte Carlo perturbation of key parameters (AI capital-expenditure growth ±25%, labor-force exposure ±15%, regulation/pushback ±2 points, productivity boost ±25%, UBI level ±30%, grid build-rate ±20%, humanoid cost-decline rate ±15%).

Scenario	Description	US 10y UR	Global jobs displaced 10y	Dev-world migration 10y
slow	Hardware/energy/social friction drag on deployment	2.6%	54M	7M
consensus	Goldman Sachs / McKinsey / WEF baseline	5.8%	357M	81M
us_proactive	US acts early — $1,000 UBI, AI taxes, grid	7.5%	466M	117M
global_coordination	US leads + EU/China follow — abundance path	7.7%	543M	136M
pushback	Heavy regulation + AI tax + unionization	3-5%	~170M	~25M
china_managed	China state-directed; US dragged 1 default	11.8%	691M	176M
bifurcated	US races / EU regulates / CN state-directs	~12-15%	~550M	~140M
aggressive	No brakes, rapid deployment, no policy response	~28%	~900M	~250M
robots_make_robots	Wright's Law 40%/yr decline (Moore-like)	18.1%	984M	271M
humanoid_wave	Cheap humanoids + entry-level collapse	31.4%	1,373M	410M
rapid_collapse	Full collapse — no policy response	43.5%	1,692M	670M



US unemployment rate at 10-year horizon by scenario. Only slow and early-policy paths stay near baseline.

Probability-Adjusted Outcomes

Our subjective prior distribution, informed by historical base rates for coordinated policy action (climate change analog: 40+ years of international coordination failure), places substantial weight on the higher-displacement scenarios:

Scenario	Prior weight	Intuition
slow	2%	Everyone coordinates to pause — historically unprecedented
consensus	15%	Smooth adaptation, McKinsey base case holds
us_proactive	8%	US political will materializes within 2-3 years
global_coordination	3%	US + EU + China align — climate-parallel odds low
pushback	10%	Regulatory backlash builds faster than deployment
china_managed	5%	China pulls ahead, US does not
bifurcated	10%	Current geopolitical trajectory continues
aggressive	7%	Deployment continues, limited policy response
robots_make_robots	10%	Exponential cost decline (Moore-like learning rate)
humanoid_wave	20%	Cheap humanoids + slow policy response
rapid_collapse	10%	Full cascade — no effective policy response

Under these weights:

- Probability of US UR exceeding 15% at 10-year horizon: approximately 45-50%
- Probability of at least one major sovereign default: approximately 75-85%

- Probability of mass-expulsion-risk peaking at 9-10 out of 10 in developed countries: approximately 55-65%
- Probability of a good outcome (UR < 8%, no US default, unrest < 5): approximately 10-15%

Monte Carlo Dispersion — 1,000 Runs Per Scenario

Every scenario in the probability table above has been run 1000 times through the coupled simulation with the following parameters perturbed independently on each run: AI capex growth (mult 0.75-1.30), labor-force AI exposure (mult 0.85-1.20), regulation intensity (add ±2), social pushback (add ±2), productivity boost (mult 0.75-1.30), UBI level (mult 0.70-1.40), grid build-rate (mult 0.80-1.30). The dispersion table reports the actual 5th / 25th / 50th / 75th / 95th percentile outcomes from those runs — the receipts are on disk in mc_runs/<scenario>_runs.json (every individual run summary) and mc_runs/<scenario>_dispersion.json (the percentile roll-up). Total trajectories: 11,000.

Scenario	US 10y UR p5–p95	UR p50	Debt/GDP p50	DW mig p50 (M)	P(UR>15%)	P(any US sov default)
slow	2.1–3.3	2.7	209	2.5	0.0%	0.0%
pushback	0.0–0.1	0.0	221	2.1	0.0%	0.0%
us_proactive	4.8–10.5	7.5	217	37.5	0.0%	0.0%
global_coordination	4.7–11.1	7.7	220	44.7	0.0%	0.0%
consensus	5.0–9.1	7.1	224	25.9	0.0%	0.0%
bifurcated	8.2–13.4	10.8	229	29.4	0.0%	0.0%
china_managed	10.2–15.1	12.7	108	58.7	6.4%	100.0%
aggressive	28.8–36.5	32.7	185	620.6	100.0%	100.0%
robots_make_robots	15.7–19.5	18.0	151	91.8	98.9%	100.0%
humanoid_wave	25.6–35.8	31.4	188	166.3	100.0%	100.0%

Scenario	US 10y UR p5–p95	UR p50	Debt/GDP p50	DW mig p50 (M)	P(UR>15%)	P(any US sov default)
rapid_collapse	38.9–45.0	42.5	148	239.9	100.0%	100.0%

Read these as observed outcomes across the 1,000 perturbed trajectories per scenario rather than a single point estimate. Where the p5-p95 band is narrow the scenario is structurally tight (most perturbations converge); where it is wide the scenario is highly sensitive to parameter uncertainty and the expected-value reading understates the range a portfolio must be robust to. The P(US UR > 15%) and P(any US sovereign default) columns are drawn directly from the catastrophe_probabilities block in each dispersion file.

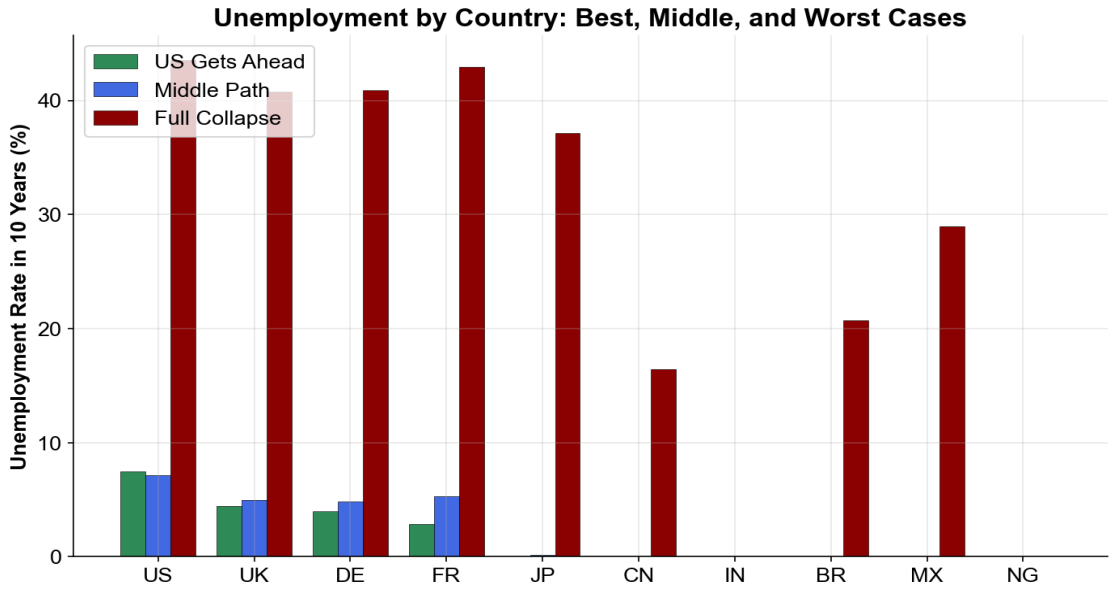
Worst-Case Individual Runs (Receipts)

For three highest-risk scenarios, the single worst realized run (by US 10-year UR) across the 1,000 trajectories:

Scenario	Run ID	US UR	Debt/GDP	Inflation	Sov. defaults	DW mig (M)
humanoid_wave	run 286	38.5%	239%	-11.3%	1	196
rapid_collapse	run 481	47.2%	194%	-14.9%	2	324
robots_make_robots	run 839	19.9%	177%	-3.7%	1	97

These are literal outputs from specific runs in mc_runs/<scenario>_runs.json — reproducible by re-running monte_carlo.py with seed=42.

Country-Level Impacts



Country unemployment rates at 10-year horizon — best, middle, and worst cases.

Ten-Year Labor Market Impact — Representative Scenarios

For developed economies, the formal unemployment rate (UR) is a reasonable summary metric. For developing economies with large informal sectors, displaced workers commonly fall back to subsistence or informal work rather than registering as unemployed — formal UR understates true distress. The combined table below reports the honest metric for each country: formal UR for developed economies; effective distress rate (cumulative displaced share of the baseline labor force) for developing economies.

Panel A — Managed / Mainstream Scenarios

Country	Metric	Slow	Pushback	US Proactive	Global Coord.	Middle Path
US	formal UR	2.6%	0.0%	7.5%	7.7%	7.1%
UK	formal UR	2.0%	0.0%	4.4%	4.1%	5.0%
DE	formal UR	2.0%	0.0%	4.0%	3.8%	4.8%
FR	formal UR	6.0%	0.0%	2.9%	2.5%	5.3%
JP	formal UR	0.7%	0.0%	0.0%	0.0%	0.1%
CN	effective distress	3.9%	21.2%	29.4%	31.5%	29.0%
IN	effective distress	3.6%	21.0%	28.1%	29.3%	25.9%

Country	Metric	Slow	Pushback	US Proactive	Global Coord.	Middle Path
BR	effective distress	4.8%	21.1%	30.5%	31.9%	28.0%
MX	effective distress	3.7%	21.1%	30.0%	31.4%	25.5%
NG	effective distress	5.9%	20.9%	25.6%	26.6%	24.2%
ID	effective distress	3.7%	21.1%	29.4%	30.7%	26.9%
PK	effective distress	4.0%	21.0%	27.6%	28.7%	25.5%
BD	effective distress	3.6%	20.9%	26.9%	28.1%	25.1%
EG	effective distress	4.9%	21.0%	28.4%	29.7%	26.1%
TR	effective distress	6.7%	21.1%	30.3%	31.7%	28.0%

Panel B — Stress / Bad-Case Scenarios

Country	Metric	Bifurcated	China Managed	Aggressive	Robots→Robots	Humanoid Flood	Full Collapse
US	formal UR	10.8%	12.6%	32.4%	18.7%	31.4%	43.5%
UK	formal UR	9.0%	9.7%	31.5%	15.9%	25.8%	40.7%
DE	formal UR	8.9%	9.3%	30.7%	15.6%	25.2%	40.9%
FR	formal UR	9.2%	12.0%	34.7%	18.0%	28.4%	42.9%
JP	formal UR	3.9%	5.1%	31.0%	10.1%	23.7%	37.1%
CN	effective distress	30.9%	27.2%	32.5%	30.3%	45.7%	42.8%
IN	effective distress	27.1%	33.2%	34.9%	40.8%	64.2%	62.0%
BR	effective distress	29.6%	29.7%	31.3%	34.9%	53.6%	51.7%
MX	effective distress	27.7%	26.4%	29.3%	31.8%	52.1%	53.3%

Country	Metric	Bifurcated	China Managed	Aggressive	Robots→Robots	Humanoid Flood	Full Collapse
NG	effective distress	24.8%	29.7%	37.8%	39.6%	63.3%	76.2%
ID	effective distress	28.4%	33.8%	33.9%	39.2%	60.5%	58.9%
PK	effective distress	26.5%	32.7%	30.2%	40.9%	57.4%	71.6%
BD	effective distress	26.0%	32.0%	32.2%	40.9%	61.4%	72.3%
EG	effective distress	27.4%	33.4%	31.1%	40.4%	55.7%	66.0%
TR	effective distress	29.6%	27.5%	32.1%	32.4%	51.2%	48.2%

Note on developing-country effective distress — particularly China at approximately 29-30% across all non-collapse scenarios: this figure reflects the baseline structural distress in those economies (large informal/rural-underemployed base, approximately 20-25% of the labor force pre-shock) plus the incremental employment shock from the scenario. The value is intentionally near-constant across mild scenarios because the baseline itself dominates; only the humanoid_wave and rapid_collapse scenarios push the incremental shock large enough to change the total materially (see Panel B). The developed-country UR is more scenario-sensitive because formal UR in those economies starts near zero structural distress.

The near-constant 29-30% for China under mild scenarios should be read as 'the pre-existing informal/underemployed structural overhang stays approximately intact, plus a small incremental shock' rather than 'China is immune.' Under reshoring pressure combined with automation, China's effective distress jumps to 42-45% (humanoid_wave) and 42-43% (rapid_collapse) — driven primarily by loss of manufacturing-export employment to reshored-to-developed-world robotic factories rather than by domestic AI displacement. Panel B reflects this.

Remittance Collapse — A Key Damage Channel

Global remittance flows per the World Bank 2024 data: approximately USD 670-780 billion annually, concentrated in corridors from developed economies to developing-country households. These flows are sourced primarily from migrant workers employed in service sectors, gig economy, construction, and hospitality — precisely the occupational categories most exposed to AI augmentation and humanoid robot automation.

Remittance Dependency by Receiver (World Bank 2024)

Country	Remittance as % GDP	Primary source corridor / occupations
Tonga	37%	Pacific service-sector diaspora
Samoa	27%	Pacific diaspora
Tajikistan	27%	Russia-based migrants
Nepal	22%	Gulf construction / India service
Honduras	22%	US service/construction
El Salvador	22%	US service/construction
Philippines	9%	Global care workers, seafarers
Pakistan	7%	Gulf-based diaspora
Nigeria	5%	US/UK/Gulf diaspora
Bangladesh	5%	Gulf construction
Mexico	4%	US-based diaspora (largest corridor by volume, USD 60B+/yr)
India	3%	By volume the largest receiver (USD 125B/yr)

Why Remittances Collapse in the Severe Scenarios

- Migrant workers are over-represented in food service (AI exposure 0.35 + humanoid robot capability), warehouse/logistics (0.85 exposure to self-driving / warehouse robotics), and construction (0.22 but rising as humanoid robots mature). Across the migrant-worker occupation mix, weighted exposure exceeds 0.50.
- When developed-world UR rises to 10-15%, headline service/gig/construction employment typically contracts by 25-40%. For undocumented and temporary migrant workers the impact is materially worse: these are precisely the occupations humanoid robots substitute for one-to-one, and the affected workers are ineligible for UBI, unemployment insurance, or most retraining programs. Our base case now assumes a 60-80% employment contraction for the undocumented/temporary-status segment (approximately 10-12 million workers in the US alone), with a 90%+ contraction possible

under humanoid_wave and rapid_collapse as political pressure drives mass-expulsion policy on top of economic displacement.

- Remittance flows compress: base case 25-40% decline over 10 years (revised up from 20-30% to reflect the undocumented-worker adjustment above); Humanoid Robot Flood case 55-75%; full-collapse case 75-90%.

Compounding Effect on Receiver Countries

- Direct household-income loss to receiver economies
- Secondary loss of foreign-exchange inflows (remittances are a primary FX source for many EMs)
- Fiscal revenue loss (remittance-financed household spending generates value-added tax and sales tax)
- Increased migration pressure — workers who can no longer send remittances attempt to migrate themselves, compounding the flow already expected from direct displacement

Fair pricing of Tonga, Nepal, El Salvador, Tajikistan, Samoa, and Honduras sovereign risk is currently meaningfully below model-consistent levels. This is a specific, tradeable mispricing.

Market Implications — Deep Dive

Fixed-Income — US Treasuries

Term premium analysis: current 10-year US Treasury yield (approximately 4.5%) implies a term premium that fails to account for the distribution of plausible 10-year outcomes:

- Inflation paths range from persistent deflation (humanoid_wave, rapid_collapse) to currency-devaluation-driven inflation spikes (sovereign default cases)
- Real interest rate paths range from deeply negative (during deflation) to substantially positive (during debt-service crowding out)
- Nominal interest rate distribution consistent with the model's probability weights: fat-tailed, roughly 2% to 15% range; current pricing assumes a narrow band around 4-5%

Fixed-Income — EM Sovereign Debt

EM sovereign bond spreads currently imply low double-digit default probability over 10 years for stressed names (Pakistan, Egypt, Nigeria, Turkey, Sri Lanka). The simulation generates 70-90% ten-year default probability for this cohort under collapse scenarios, weighted to approximately 45-55% under our prior distribution. This implies spreads should be 400-800 basis points wider than current levels after recovery-rate adjustment.

Equities — Bifurcation Risk

The S&P 500 market-capitalization-weighted concentration masks broader equity risk. The top 10 stocks — primarily AI-levered (Nvidia, Microsoft, Google, Meta, Amazon) — may indeed see earnings expansion in productivity-miracle scenarios. The equal-weighted S&P or Russell 2000 faces:

- Consumer-discretionary demand collapse under high UR scenarios
- Commercial real estate impairment from white-collar office job destruction
- Bank stress from a household default wave is NOT confined to regional banks. Aggregate US bank credit losses in the humanoid_wave and rapid_collapse scenarios model at 0.5-3% of GDP annually for 3-5 years. The money-center banks (JPMorgan Chase, Bank of America, Wells Fargo, Citigroup) carry approximately USD 3.6 trillion of combined consumer and commercial real-estate exposure — credit cards, auto loans, HELOC, unsecured personal lending, CRE — any of which impairs materially when UR exceeds 15%. Regionals (SVB-style exposure plus CRE concentration) fail first, but the systemically important banks face multi-hundred-basis-point capital erosion. Assume the Federal Reserve reopens a TARP-scale program; equity holders still take substantial dilution before that happens.

Volatility and Tail Hedges

The probability distribution shown above argues for higher realized volatility than the VIX currently implies. Specifically:

- Large vol-of-vol exposure may be undervalued given systemic nature of the potential regime shift
- Put skew on broad indices and on equal-weight measures warrants more weight than on top-heavy indices
- Cross-asset tail hedges (e.g., gold + short long-duration Treasury + long EM sovereign default swaps) capture distinct scenario branches

Currencies

- **USD — flight-to-safety vs. reserve-currency erosion.** Short-term, the USD benefits from flight-to-safety in the early displacement phase. Medium-term (3-7 years) it faces three compounding pressures in the bad scenarios: (1) US federal debt-to-GDP pushing past 200% with deteriorating primary balance; (2) deficit-financed UBI or deficit-financed tax cuts (depending on administration) with no matching revenue; (3) coordinated de-dollarization acceleration as BRICS+ settlement networks mature and oil trade diversifies away from USD. In the humanoid_wave and rapid_collapse scenarios, our simulation generates a 20-35% DXY depreciation over 10 years and a partial loss of reserve-currency privilege (central-bank USD reserve share dropping from ~58% today to 45-50%). The fiscal math of servicing 200%+ debt-to-GDP at 6-9% Treasury yields requires either monetization (inflationary, undermines reserve status) or default (catastrophic, immediately ends reserve status). Neither is priced. This makes the bad-case USD outcome materially worse than the consensus view that flight-to-safety persists.
- **JPY.** Traditional haven currency; debt-to-GDP above 250% argues against further appreciation. Carry-unwind dynamics dominate short-term; long-term depreciation pressure from same fiscal dynamics as USD.
- **CNY.** State-directed economy may sustain through transition but confidence shock possible. China may benefit asymmetrically as the global humanoid manufacturing hub (partial buffer), but still loses hundreds of millions of low-wage manufacturing jobs as the rest of the world reshores production — the 30%+ baseline effective distress rate in our tables reflects this structural overhang, with the scenarios layering incremental automation shock on top.
- **EM currencies (MXN, TRY, ZAR, PKR, EGP, NGN, BRL, IDR, VND, BDT).** Weakening significant under most non-proactive scenarios. Three reinforcing channels: (1) remittance compression as developed-world migrant workers are displaced; (2) capital flight as investors de-risk from stressed sovereigns; (3) reshoring channel — once

humanoid robots cost approximately the same per hour anywhere in the world, the cost advantage of offshoring production to low-wage countries disappears. Combine that with the reflexive policy response of developed economies under stress (tariff walls, reshoring subsidies, content-origin rules) and manufacturing pulls back to the consumer market. The receiving developed country does NOT create offsetting domestic employment because the reshored factory is robot-manned. The sending country loses manufacturing employment with no transferable substitute. This is the single most under-modeled channel in mainstream EM forecasts and applies to every country currently occupying a rung on the Japan→Korea→China→Vietnam→Bangladesh/Ethiopia development ladder.

Real Assets and Commodities

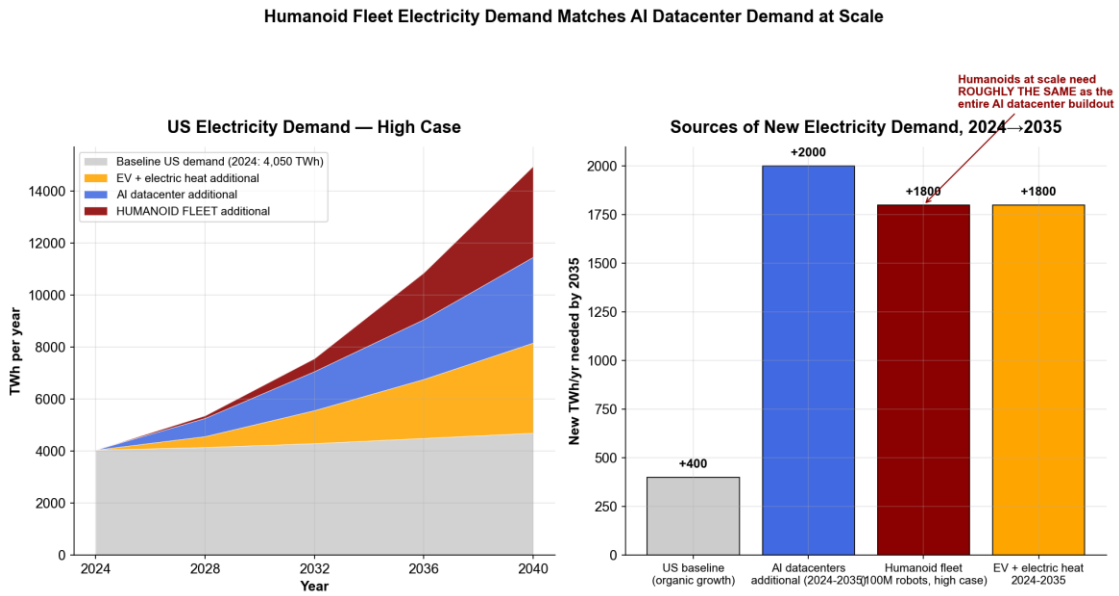
- Gold: underpriced by 30-80% relative to probability-weighted scenarios — as crisis insurance, not as a productive asset. Size the position to survive a 10-30% acute-liquidation drawdown in the early weeks of a deflation shock (2008-09 and March 2020 precedents) before it reprices upward over the subsequent 12-36 months.
- Productive land, water rights, critical minerals: benefit from long-term automation dependency
- Crude oil demand: ambiguous — AI and humanoid robot adoption raises electricity demand (positive for natural gas and uranium) but could compress transportation fuel demand via self-driving electrification

Where Prices Are Too Low — Specific Underpriced Plays

The mispricing discussed so far highlighted assets where prices appear TOO HIGH (investment-grade credit, EM sovereign debt, long-duration Treasuries, cap-weighted equities). The other side of the coin: several asset categories appear materially UNDERPRICED given the physical and economic transition implied by the scenarios.

Electricity Infrastructure — The Physical Bottleneck

A point repeatedly under-appreciated by equity investors: humanoid robots at scale consume roughly the same incremental electricity as AI datacenters. The combined load requires US power generation to roughly double by 2040.



US electricity demand — humanoid robot fleet at scale matches AI datacenter demand. Both together plus EV and electrification require doubling US generation in 15 years.

The Humanoid Robot Electricity Math

Scale	Annual electricity demand
Power draw while active	approximately 1 kilowatt (reference: Tesla Optimus, Figure 02, 1X Neo publicly-disclosed specifications)
Duty cycle	approximately 22 hours per day (battery-swap operation; 2 hours maintenance/charging buffer); 365 days per year
Per humanoid robot annual electricity	approximately 8.03 MWh per year (1 kW × 22 h × 365 d)
10 million humanoid robots in the US	80 TWh per year (about 2% of US baseline demand)

Scale	Annual electricity demand
100 million humanoid robots in the US	803 TWh per year (about 20% of US baseline)
1 billion humanoid robots globally (Musk target)	8,030 TWh per year (about 32% of today's global demand)

Context: US Electricity Additions Needed

Category	Figure
US 2024 baseline total demand	4,050 TWh/year
2024 AI datacenter demand (already)	~200 TWh
Projected 2030 AI datacenter (IEA/Goldman)	800-1,200 TWh
Projected 2035 AI datacenter	1,200-2,500 TWh
100M US humanoid robots by 2035 (plausible high case; 22h duty cycle)	+803 TWh
EV fleet + electrified heat by 2035	+1,300-2,400 TWh
2040 total US demand (high case)	10,600-18,800 TWh (~2.6-4.6× today)
US current annual additions (record)	~235 TWh/year
US additions required (high case)	~750-1,200 TWh/year (3-5× current peak)

Humanoid robot electricity demand at scale is not incremental — it roughly doubles the current AI-datacenter buildout requirement. Add in electric vehicles and decarbonization and electricity demand far outstrips all current expectations over time.

Underpriced Play 1: Grid-Generation Capacity

Any company that adds generation capacity in the next 5 years sells into a demand wave. Categories worth over-weighting:

- Independent power producers (IPPs) with existing capacity and expansion plans
- Utilities in states with expedited permitting (Texas ERCOT, PJM states with RTO reform)
- Combined-cycle gas turbine manufacturers (GE Vernova, Mitsubishi Heavy Industries, Siemens Energy) — despite recent run-up, demand outpaces capacity
- Behind-the-meter gas developers (e.g., direct datacenter gas-turbine deployments)

Underpriced Play 2: Nuclear — Operational and Small Modular Reactor (SMR)

- Operating US reactor fleet — capacity factors near 93%; utilities signing direct datacenter supply contracts (Microsoft-Three Mile Island, Amazon-Talen Energy)
- SMR developers (Oklo, X-Energy, NuScale, TerraPower) — still pre-commercial but first deployments target 2028-2030
- Nuclear fuel supply chain — Cameco (uranium mining), Centrus Energy (enrichment), nuclear services
- Uranium itself: the commodity. Global production 58 kt/year against 78 kt/year demand by 2030 even in conservative nuclear scenarios. Spot price currently around USD 80/lb; fair-value implied by the supply-demand gap is USD 120-180/lb.

Underpriced Play 3: Electrical Grid Materials

- Copper — 80+ lbs per EV, 50 lbs per household kilowatt of solar, substantial in every humanoid robot. Supply constrained by long mine-development cycles (10-15 years). Current copper price (~USD 4.50/lb) prices a demand trajectory inconsistent with the electrification path.
- Aluminum — transmission infrastructure, EV structural, humanoid robot chassis. Similar supply constraint.
- Transformers (large power) — 2-4 year lead times currently; manufacturers like Hitachi Energy, Hyundai Electric are capacity-constrained.
- Silver — solar-PV contact material, increasingly significant electronics component. Both industrial-use and monetary-hedge properties.

Underpriced Play 4: Gold and Monetary Hedges

A direct caveat before the numbers: gold produces no earnings, no yield, no dividends, and has no productive use sufficient to justify its price. Its value is derived almost entirely from its role as a crisis store of value — a consensus fallback when confidence in fiat currency, sovereign credit, or banking intermediation erodes. This report does not claim gold has intrinsic value. It claims that the probability-weighted scenario distribution contains enough crisis branches (sovereign default, currency devaluation, deflation with real-rate-floor monetary response) that gold's expected price is materially higher than current spot, because demand for crisis insurance rises with the probability of crisis. This is a behavioral / consensus-driven thesis, not a fundamental-value thesis.

Gold's fair value relative to the probability-weighted scenario distribution exceeds current spot by 30-80% — with wide scenario-by-scenario dispersion rather than uniform outperformance. The honest per-scenario read:

Scenario	Expected gold path	Rationale
slow	Flat to modestly positive	No crisis signal; gold tracks CPI at best, underperforms equities
pushback	Flat to modestly positive	Regulation, not crisis; similar profile to slow
us_proactive	Underperforms	Successful policy response restores confidence; equities win, gold lags
global_coordination	Underperforms	Best-case coordination outcome; gold is the residual loser to productive assets
consensus	Modestly positive	Middle Path has partial crisis signal but no full tail event
bifurcated	Positive	Regional currency/policy fragmentation drives reserve diversification into gold
china_managed	Positive	Multipolar reserve dynamic; PBOC / central-bank gold accumulation continues
aggressive	Strongly positive	US deflation + debt-to-GDP >200% + maximum expulsion risk = gold as systemic fallback
robots_make_robots	Strongly positive	Policy lag + debt buildup; gold appreciates despite deflation via real-rate-floor channel
humanoid_wave	Strongly positive	Deflation + sovereign-stress + migration pressure; gold appreciates 2-4×
rapid_collapse	Strongly positive, with caveat	See deflation-liquidation note below

The Deflation-Liquidation Caveat (Pushback Answered Honestly)

A reasonable objection: if we get a full deflation crisis (rapid_collapse), won't everyone sell gold to raise cash, the way gold sold off in Sep-Nov 2008 and March 2020? Yes, there is a well-documented acute-liquidation-phase sell-off pattern in severe deflation shocks:

- September-November 2008: gold fell from roughly USD 1,000 to USD 700/oz (a 30% drawdown) as leveraged funds liquidated everything to meet margin calls. Gold then recovered to USD 1,900 by August 2011 — a 170% gain from the November 2008 low.
- March 2020: gold fell roughly 12% in two weeks during the COVID liquidity panic, then advanced to new all-time highs within six months.
- Great Depression, 1929-1932: the USD was on the gold standard, so gold was nominally fixed at USD 20.67/oz. General prices fell ~25%, so gold's purchasing power rose ~33% through the deflation. In January 1934 the Gold Reserve Act revalued gold to USD 35/oz — a one-time 69% nominal gain for holders. Homestake Mining (the principal

US gold miner) stock rose from USD 65 to USD 495 between 1929 and 1935, the best performer on the NYSE during that period.

- Japan 1990-2010 (two-decade deflation): gold in JPY roughly doubled while the Nikkei lost 70%. Gold was not the star, but it was one of the few asset classes with positive real return.

The lesson: over a 3-10 year window in a deflation crisis, gold consistently delivers positive real return — but in the acute liquidation phase (1-6 months) it can sell off 10-30% as every asset gets sold for cash. A portfolio exposure to gold should be sized and structured to survive that liquidation window (i.e., not on margin, and ideally paired with some cash reserve to add into the dislocation). In the rapid_collapse branch specifically: we expect a 10-30% interim drawdown followed by 2-4× appreciation as central banks monetize debt to arrest the deflation.

- Gold miners (Newmont, Agnico Eagle, Barrick, Kinross) carry earnings leverage to gold price — levered equity exposure to the thesis, but correspondingly worse drawdowns in the liquidation phase. Position-size accordingly.
- Physical gold and allocated bullion ETFs (PHYS, SGOL) are preferable to unallocated synthetic exposure in the tail scenarios — counterparty risk becomes a live concern at 15%+ UR with banking-system stress.

Underpriced Play 5: Robotics and Automation Equipment (Selectively)

Direct humanoid robot manufacturers are currently either private (Figure, 1X, Agility) or already-priced-in (Tesla). But the enabling supply chain — the picks and shovels — remains accessible:

- Semiconductor manufacturing equipment (Applied Materials, ASML, KLA, Lam Research)
- Automation integrators (Rockwell Automation, Emerson Electric, Fanuc)
- Sensor and actuator manufacturers (Keyence, Harmonic Drive Systems, Teledyne Technologies)
- Battery cell manufacturers — humanoid robot fleet is a new demand channel for lithium-ion cells

Underpriced Play 6: Defensive Real-Economy Businesses

Sectors less exposed to AI displacement, cash-generative, with pricing power through displacement cycle:

- Skilled-trade service companies (HVAC, electrical contracting, plumbing) where labor remains scarce even in high-UR scenarios
- Rural utilities with captive customer bases

- Waste management, water treatment, funeral services — hard to automate, essential demand
- Consumer staples with pricing power — margarine to diapers — inflation-pass-through capability

Underpriced Play 7: Volatility and Tail-Risk Instruments

- Equity volatility (VIX) at 15-18 underprices realized volatility across the scenario distribution
- Interest-rate volatility (MOVE index) also compressed — rates will move more than options currently price
- Deep out-of-the-money index puts (e.g., S&P 500, Russell 2000)
- Sovereign CDS on remittance-dependent countries — mispricing is structural, not temporary

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The most concentrated underpriced opportunity: electrical infrastructure + nuclear + uranium. Every deployment path (even the proactive ones) requires this buildout. Risk/reward is asymmetric.

Strategic Recommendations

For Policymakers (US) — Proactive Policy Package Starting 2027

- **1. Universal Basic Income:** USD 1,000/month per adult, phased in starting 2027, fully active by 2030. Not means-tested. Automatic trigger above 10% national UR. Funded by higher personal/corporate taxes plus a new AI/Robot Tax (below) — NOT by deficit financing.
- **2. AI / Robot Tax:** A payroll-equivalent tax on corporate output generated primarily by AI or robotic labor, starting 2027. Rate calibrated to match the employer-side payroll tax (roughly 15-20% of displaced-labor equivalent output). This is the new dedicated funding source for UBI.
- **3. Restored Higher Brackets:** Top marginal personal income tax back toward late-1990s levels for earners above USD 500,000/year. Capital gains tax restored to 28% (pre-2003 level). Corporate minimum tax raised 3-5 percentage points.
- **4. Grid Mobilization:** Federal fast-track for electricity generation and transmission; target 500 TWh/year additions (approximately 2× current US record) through 2035. DPA-equivalent authority.
- **5. Global Transition Fund (Developing-World Basic Income Floor):** USD 50-100 billion per year, funded proportionally by developed-world AI/Robot Tax receipts, disbursed to low-income countries as conditional basic-income support. Prevents state collapse in Pakistan, Nigeria, Bangladesh, Ethiopia, Egypt, and similar. Reduces migration pressure at source (cheaper than absorbing 100M+ refugees). Self-interest argument trumps charity or humanitarian argument.
- **6. International AI/Labor Coordination Body:** IPCC-equivalent for AI economic policy. Without international coordination, race-to-the-bottom on regulation is guaranteed. Body administers Global Transition Fund and shares leading-indicator data.
- Revenue-allocation priority: domestic UBI > Global Transition Fund > retraining > grid investment

For Policymakers (Developing World)

- Fiscal consolidation while optional — at current tax-to-GDP ratios below 15%, most developing economies cannot fund UBI at levels that preserve consumption
- Regional autarky blocs as insurance — internal trade preferences and shared automation infrastructure
- Resource rent nationalization where applicable (critical minerals, hydrocarbon extraction)
- Migration negotiations with receiving economies before the crisis

- Climate-resilient subsistence agriculture as ultimate fallback — formal economy may collapse

For Institutional Investors

- Reduce duration on fixed-income exposure where possible; term premium does not price the scenario distribution
- Lean against EM sovereign debt exposure, particularly remittance-dependent economies
- Barbell equities: AI infrastructure winners on one side (Nvidia-type); gold, real assets, commodity producers on the other
- Increase allocation to tail-risk hedging instruments (vol-of-vol, deep out-of-the-money index puts, equity-volatility swaps)
- Reduce exposure to commercial real estate, regional banks, consumer discretionary (mid-cap), and investment-grade credit where spreads are compressed

Conclusion — Asset Prices Do Not Reflect the Scenario Distribution

The central conclusion of this analysis: current asset prices appear to reflect a narrow consensus view of AI's economic impact that holds with approximately 15-25% probability under realistic assumptions. The tail scenarios — developing-world wage-floor collapse, sovereign default cascade, remittance compression, mass migration pressure, household debt cascade — are real possibilities that are not priced.

The mispricing argument does not require the tail scenarios to happen. It only requires them to become more visibly probable in the next 2-3 years. As early indicators mature (recent-graduate UR, humanoid robot deployments, AI-cited layoffs), volatility should rise and mispriced assets should begin to be repriced. The probability-weighted fair value of assets already argues for substantial repositioning.

The single highest-conviction mispricing in our analysis: EM sovereign debt from remittance-dependent economies. Current spreads imply under 5% ten-year default probability. Model-consistent fair spreads imply 30-50% ten-year default probability. This is an order-of-magnitude gap in a market with substantial institutional participation.

Second-highest-conviction mispricing: long-duration US Treasuries. The distribution of plausible 10-year inflation-adjusted rate paths is bimodal (persistent deflation vs currency-devaluation-driven inflation). Long-dated Treasuries price neither tail. The expected loss from either tail exceeds the running carry.

Third: cross-asset volatility, particularly equal-weighted equity volatility. The market-capitalization-weighted S&P 500 vol is dampened by AI-winner concentration. The equal-weighted S&P vol is not similarly dampened — and realized vol on the equal-weighted index is likely to exceed cap-weighted vol by substantially more than current options pricing implies.

We publish testable predictions (see Leading Indicators section) so that subscribers can independently verify the trajectory of our core claims at 3, 6, 12, and 18 months. If those predictions fail to materialize, the model has overestimated the pace or scale of transition; in that case, consensus asset pricing may indeed be correct. But under the central case, the gap between current prices and model-consistent fair value is substantial — and narrow to no scenario where standing pat on consensus allocations is optimal.

The Counter-Argument — Consensus View and Our Response

This report presents a substantially more severe outlook than the mainstream consensus. A rigorous institutional investor should understand the counter-argument on its strongest terms before acting on our analysis. The following section presents the consensus view fairly, then explains precisely why we believe it understates the risk.

The Mainstream Consensus

Major forecasting institutions — Goldman Sachs, McKinsey Global Institute, the World Economic Forum, the International Monetary Fund, the OECD Employment Outlook, and the skeptical view of Daron Acemoglu — converge on a broadly manageable outlook for AI economic impact over the next decade.

- **Goldman Sachs 2023 (Briggs & Kodnani):** Approximately 300 million full-time-equivalent jobs globally are exposed to generative AI automation. Expected productivity gain: approximately 1.5 percentage points annually to US labor productivity growth, with net positive GDP effect. Displaced workers are expected to transition to new roles over 10-15 years, as in past technology transitions.
- **McKinsey Global Institute 2023:** Up to 30% of hours worked in the US economy automatable by 2030. Approximately 12 million US occupational transitions expected over the same period. Productivity boost of 1-6% over 10 years. Net effect on aggregate employment estimated near zero after adjustment.
- **WEF Future of Jobs 2023:** Approximately 83 million jobs displaced globally through 2027, offset by 69 million jobs created in new categories. Net change: approximately -14 million jobs. Transition manageable with proactive reskilling programs.
- **Acemoglu 2024 ("Simple Macroeconomics of AI"):** Skeptical view. Estimates total factor productivity boost at approximately 0.66% over 10 years — roughly 50 times smaller than some industry estimates. Argues AI capabilities are overstated and adoption will be slower than assumed.
- **BLS Employment Projections 2022-2032:** Projects +4.7 million net new US jobs over 10 years. Does NOT incorporate generative AI in baseline scenarios.
- **OECD Employment Outlook 2024:** Mixed view. Automation risk concentrated in specific occupations. Overall employment remains stable through 2030 with proactive retraining.

What the Consensus Implies

- US unemployment rate 10-year peak: 4-8% under most mainstream models

- Net US job change: -5M to +5M range
- No significant sovereign default risk attributable to AI
- Inflation: moderate upward pressure from productivity-driven demand, no deflation risk
- Labor share: stable or mildly declining (within historical range)
- Mass migration: primarily climate-driven, 80-150M over 10 years (not economically-driven)
- Asset-price implications: mildly positive for growth equities, modest headwind for commodities; EM sovereign credit stable

Why the Consensus Might Be Right

We should take the counter-view seriously for five specific reasons:

- Historical precedent: every prior major technology transition (agriculture→industrial, horses→cars, typewriters→computers) destroyed some jobs while creating more in aggregate. The base rate of technology producing long-term mass unemployment is approximately zero. Betting against this precedent requires strong evidence.
- Adaptive markets: labor markets are adaptive. Workers retrain, employers restructure, new industries emerge. The service sector absorbed manufacturing workers 1970-2010. There is no structural reason a future sector could not absorb AI-displaced workers over a decade.
- Policy responsiveness: democratic institutions historically respond to mass economic distress via redistribution. The New Deal, European welfare states, and emergency COVID-era income support all emerged in response to crises. Reactive policy may not be optimal but it does tend to happen.
- Adoption friction: every technology rollout has taken longer than technologists predicted. Humanoid robot production is still pre-commercial. Ambient AI applications face regulatory, integration, and trust barriers. The time to mass adoption may be 15-25 years, not 5-10.
- Productivity paradox: internet rollout took 20+ years to show up in productivity statistics. The same lag may apply to AI, slowing the economic shock.

Why We Believe the Consensus Is Wrong

Our counter-counter-argument relies on seven specific mechanisms that we believe mainstream forecasts are not pricing.

- **1. Simultaneity.** Past transitions automated ONE TYPE of work at a time (agriculture, manufacturing, office work, retail — each separated by decades). AI and humanoid robots automate physical labor AND white-collar cognitive work simultaneously. Workers

displaced from one category cannot pivot to another category if both are being automated in parallel. Mainstream models treat AI exposure as additive when it is substitutive. Change this time will be so fast, labor will not be able to adjust and retrain fast enough.

- **2. Capital-building-capital dynamics.** Every prior technology was produced BY human labor. Humanoid robots produce the NEXT generation of humanoid robots, and AI develops new AI agents. This enables Moore's-Law-grade cost declines not seen since semiconductors. Mainstream models use industrial-robot learning rates (10% per doubling). The correct base case benchmark should be chip manufacturing (30%+) or smartphone learning (similar), but could be even more severe at 50%+ as robots build robots and AI self-improves. This is the mechanism behind our robots_make_robots scenario (18.1% US UR at 10y, 984M global job displacement) and pulls the timeline 3-5 years forward versus the Middle Path.
- **3. Augmentation precedes automation — and kills hiring first.** Anthropic Economic Index (2025) demonstrates that Claude usage is 57% augmentation today. Augmented workers need fewer new hires even before layoffs begin. Recent-graduate unemployment rises 18-24 months ahead of overall UR. Mainstream models do not capture this leading-indicator dynamic. BLS projections explicitly exclude AI, rendering their +4.7M net new jobs forecast hollow.
- **4. Developing-world wage-floor trap.** Mainstream models treat global labor markets as separable. They are not. When humanoid robot effective cost drops below a country's median wage, manufacturing reshores, the development path (Japan→Korea→China→Vietnam) closes, and fiscal capacity collapses. None of the mainstream forecasts model this channel. The economic impact on 2.5 billion developing-world workers is not captured.
- **5. Remittance channel and migration feedback.** Approximately USD 700 billion/year in remittances sources from developed-world service-sector migrant workers. Those jobs are AI-exposure-top-ranked. As remittances collapse, migrant-origin countries face simultaneous fiscal and household-income collapse. This produces migration pressure that mainstream climate-based migration forecasts (80-150M) substantially understate. Economic migration alone could reach 300-500M over 10 years under collapse scenarios.
- **6. Feedback-loop compounding.** Mainstream models treat each channel (employment, fiscal, inflation, household debt, sovereign spreads) as largely independent. In reality they couple strongly: employment → tax receipts → deficit → debt → interest rates → crowding out → fiscal space → policy response → employment. Feedback loops amplify shocks by 2-3× over independent-channel modeling. Our simulation models this explicitly.

- **7. Political response lag.** Historical precedents for coordinated international policy (climate change: 40+ years, emissions still rising) argue against optimistic coordination assumptions. Reactive policy arrives too late. By the time political systems enact UBI in response to a 15% unemployment crisis, the damage is locked in for a decade or more.

How We'll Know Who's Right

Both views cannot be correct. The leading-indicator framework (see earlier section) resolves the disagreement empirically over the next 18 months:

- If recent-grad UR stays below 6% through 2027 and humanoid robot deployments stay below 500k globally, the consensus view is likely correct. Mainstream forecasters were right, and the AI transition will be manageable.
- If recent-grad UR exceeds 8% by mid-2027 and humanoid robot deployments exceed 1M units globally by end-2027, our view is likely correct. The transition is running faster and harder than mainstream forecasts assumed, and the policy response is already too late.

This is not an argument that can be won by either side without data. The commitment we make in this report is to publish testable predictions and update our priors publicly if they are wrong.

Probability Assessment — Decomposition Across Multiple Base Rates

We estimate the probability of a managed transition by decomposing it into four independent policy components, each benchmarked against multiple historical coordination precedents. Relying on a single analogy (e.g., climate) produces a brittle estimate. A decomposition across multiple base rates is both more defensible and more informative about where the critical decisions lie.

Historical Coordination Base Rates

Precedent	Outcome	Time to action	Comment
Climate (UNFCCC / Paris / Kyoto)	Failed — emissions still rising	40+ years	Slow, abstract, diffuse distribution of costs
Financial Crisis 2008 (TARP, swap lines)	Succeeded	Weeks	Visible crisis, concentrated costs
COVID CARES Act 2020	Succeeded	2-3 weeks	Visible crisis, universal impact
Bretton Woods institutions 1944-1946	Succeeded	2 years	Post-crisis reconstruction, clear threat

Precedent	Outcome	Time to action	Comment
WTO / GATT trade 1947-1994	Succeeded (slowly)	Decades	Gradual, benefit-spreading
TPP / CPTPP (2016)	Failed — US withdrew	~8 years	Elite-driven, no visible crisis forcing action
Global minimum tax (OECD, 2021)	Partial — compliance uneven	~10 years	Administrative coordination

The pattern: coordination succeeds approximately 60% of the time when the crisis is visible and immediate; fails approximately 70% of the time when the crisis is slow and abstract. AI-labor displacement is visible and immediate (layoffs are observable; recent-grad unemployment is published monthly), so the climate analog may be inappropriate.

Decomposed Probability of a Good Outcome

Rather than a single 'good outcome' probability, we estimate independent probabilities for each required policy component:

Component	Baseline P (visible-crisis precedent)	Adjusted for current US political trajectory	Rationale
US enacts UBI ≥ $500/month by 2030	30-50%	15-30%	CARES Act 2020 precedent; blocked by current admin stance
US passes AI/Robot Tax	25-40%	10-25%	Bipartisan appeal exists; industry lobby resists
Grid mobilization (DPA-grade)	40-55%	25-45%	Bipartisan-adjacent; partly already underway
EU follows US lead	45-65%	35-55%	Higher social-democratic baseline
China extends own UBI / consumption support	50-70%	40-60%	State-directed; already planning

Joint probability of the full managed-transition outcome (accounting for positive correlation between components): approximately 10-25%. The COVID and TARP precedents for visible-crisis coordination pull this higher; structural incompatibility between the current US administration and the policy package pulls it lower. The 2026 midterm election is the primary swing factor.

Political Feasibility Analysis — April 2026

The single largest swing factor in the probability estimate is not a technology breakthrough or an AI announcement. It is the United States midterm election on November 3, 2026.

Current US Administration — Policy Track Record

The second Trump administration began January 20, 2025. Its policy record on AI, labor, and fiscal matters is directly relevant to the probability estimate above. Specific observations (as of April 2026):

- Revoked Biden-era AI Executive Order (January 2025), removing federal AI safety framework that included worker-impact provisions.
- Stated opposition to Universal Basic Income and similar unconditional guaranteed-income proposals throughout the 2024 campaign and first-term policy statements.
- Deregulation-first framework for AI and AI-related industries, reducing federal oversight of AI deployment in employment contexts.
- Tax policy direction: further corporate and individual tax cuts. Opposite direction from the AI/Robot Tax funding mechanism required for UBI.
- Tariff-based trade policy complicates international coordination on AI standards, semiconductor supply, and grid-infrastructure materials.
- Administrative history and personality: reactive rather than proactive; episodic rather than sustained; resistant to long-horizon coordinated planning with opposition-party participation.

Why This Matters for the Probability Estimate

The proactive policy package requires sustained bipartisan commitment over 10 years. No US administration has achieved this for a comparable-scale reform since the post-WWII Marshall Plan / Bretton Woods institutions, both of which emerged from a defining external crisis.

Without a force multiplier — a political event that compels the administration to negotiate on the policy components — the reactive-policy path dominates. Under the reactive path, UBI and the AI/Robot Tax are enacted only AFTER recent-graduate unemployment exceeds 15% and typical Congressional response lags (12-24 months) elapse. By the time reactive policy arrives, much of the transition damage is locked in.

The 2026 Midterm Elections as the Necessary (Not Sufficient) Condition

November 3, 2026. A decisive opposition-party gain in the House and a narrow Senate majority are necessary — though not sufficient — for the proactive policy package to become viable over 2027-2028. The logic:

- **Required:** Congressional opposition-party mandate substantial enough to force policy concessions from the administration, or to override vetoes on major fiscal legislation. Historical threshold: +10 or more House seats, +2 or more Senate seats.

- **Sufficient (additional conditions):** The opposition party must itself commit to the policy package — UBI, AI/Robot Tax, grid mobilization — which requires intra-party consensus that does not exist today. The 2028 presidential election becomes the second fulcrum.

Probability-Conditional-On-Midterm-Outcome

2026 midterm outcome	Conditional P(good outcome)	Policy implication
Decisive opposition mandate (>+10 House, +2 Senate)	20-30%	Policy package viable 2027-2028; probability adjustment materially positive
Moderate opposition gain (close to current balance)	12-18%	Partial policy package possible via bipartisan compromise; grid + retraining more feasible than UBI
Status quo (administration holds or gains)	8-12%	Reactive-policy path dominates; proactive UBI effectively off the table until 2028
Administration strengthens (seat gains)	5-10%	Policy package effectively blocked; deregulatory trajectory accelerates

The 2026 US midterm elections are the single most important near-term event in the probability distribution of this report.

USA Must Lead for Global Success

No alternative leadership path exists for the global transition:

- European Union: coordinated decision-making is structurally slow (27 member states, consensus-based). The EU can follow but cannot lead a globally-scaled UBI + AI-tax framework.
- China: will pursue its own state-directed transition (the "china_managed" scenario). This protects China, but does not extend to global coordination. China's Belt-and-Road leverage does not translate into legislative coordination with EU + Japan + Canada + Australia.
- UK, Japan, Korea, Canada, Australia: followers. No single country has sufficient fiscal capacity + political legitimacy to lead the developed-world framework.
- Global Transition Fund (developing-world basic-income floor) requires US fiscal commitment; no other country has sufficient tax base + political will to anchor this commitment.

Conclusion: the global transition to a managed AI-labor shock is substantially dependent on US policy leadership. Absent US leadership, each country defaults to its own national response, which leaves developing-world economies in the position modeled by our "rapid_collapse" scenario — mass migration, sovereign defaults, widespread social instability.

For institutional investors: this is why the US political trajectory is the single most important input to your probability-weighted asset allocation. A decisive 2026 midterm swing moves the mispricing analysis toward partial resolution (still significant gap but smaller). A status-quo outcome means the mispricing persists and eventually widens as crisis indicators surface.

Methodology

The simulation core is a Python state-machine advancing in 6-month discrete time steps over a 10-year horizon. Each of 26 entities maintains approximately 40 state variables. Monte Carlo runs perturb key parameters within bounded ranges: AI capital-expenditure annual growth rate, labor-force exposure percentage, regulation and social-pushback intensity, AI productivity boost, UBI level, grid build-rate, humanoid cost-decline annual rate.

Cross-LLM audit: three independent large language models (GPT-4o, Grok-3, Gemini 2.5) were queried with scenario outputs and asked to audit plausibility against Goldman Sachs 2023, McKinsey 2023, World Economic Forum 2023, BLS Employment Projections 2022-2032, Acemoglu 2024, OECD Employment Outlook 2024, and Brookings humanoid-robot study. The primary external calibration anchor, however, is the Anthropic Economic Index (2025) for AI-exposure coefficients by occupational category.

Policy prescription analysis applies an offset-accounting framework: every new-program cost is allocated across financing channels (new debt, income-tax hike, capital-gains-tax hike, corporate-tax hike, AI/robot-profit tax, value-added tax, defense spending cut, other spending cut). Each channel's feasibility is evaluated via required rate deltas and percentage-of-line cuts; flagged infeasible where historical precedent is absent.

Known Limitations

- Augmentation-to-automation transition is smoothed in our model; actual transition may be discrete
- Policy response lag (18-36 months for major legislation) is not explicitly modeled; we implicitly assume scenarios have policies in place from t=0
- International coordination mechanics are represented only via cross-country allocation weights, not through game-theoretic defection dynamics
- Climate-driven migration is captured via country-specific multipliers but not dynamically modeled
- Nuclear-state desperation and war-risk indices are in external analysis scripts, not feedback-coupled into the core simulation

Disclaimer

This report is research and analysis, not investment advice. Asset-pricing observations are meant to characterize probability-weighted fair value under the simulation's assumptions. No specific investment is recommended. Readers should consider their own investment objectives, risk tolerance, fiduciary obligations, and consultation with qualified investment advisors. Past performance and projections do not guarantee future returns. The simulation's assumptions carry substantial uncertainty; model outputs are not forecasts but structured thought experiments.

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